ENGINEERING SIMULATION & TECHNICAL REPORT

HYDROPOWER ENERGY GENERATION AND PLANT PERFORMANCE ANALYSIS

Python-Based Hydropower System Simulation & Investigation of Design Parameters

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1. INTRODUCTION

Hydroelectric power is one of the renewable energy generation methods based on the principle of converting the potential energy of water into mechanical energy and subsequently into electrical energy. The amount of energy generated in hydroelectric power plants depends on various parameters such as water flow rate, head height, hydraulic losses, and the efficiency of turbine-generator systems.

To evaluate the performance of a hydroelectric power plant, calculating only the theoretical energy production potential is not sufficient. The characteristics of the flow reaching the turbine from the intake structure, friction losses in penstocks, environmental flow releases, and turbine capacity must also be taken into account.

Within the scope of this project, a simulation model representing a hydroelectric energy generation system was developed using the Python programming language. Through this model, the effect of different hydraulic conditions on energy generation was examined, and the impacts of specified design parameters on system performance were evaluated.

The primary objective of the study is to practically examine the mathematical foundations of hydroelectric energy generation, evaluate the impact of hydraulic losses, and analyze the energy generation amounts obtainable under different operating conditions.

2. OBJECTIVE AND SCOPE OF THE PROJECT

The main purpose of the project is to examine the key parameters affecting hydroelectric energy generation through mathematical calculations and computer-aided simulation. In this regard, the following studies were carried out:

- Investigation of the effect of water inflow rate on energy generation.
- Evaluation of the impact of ecological flow release on the flow rate downloadable to the turbine.
- Calculation of hydraulic head losses occurring in penstocks.
- Examination of the relationship between gross head and net head.
- Analysis of the effect of turbine flow capacity on energy generation.
- Calculation of monthly energy generation amounts.
- Comparison of different design parameters.
- Visualization of the obtained results through charts and tables.

The project is not a comprehensive digital twin containing all physical and electromechanical components of a real hydroelectric power plant. It is an educational and engineering analysis-oriented simulation study created to examine basic hydraulic and energy generation relationships.

3. THEORETICAL FRAMEWORK AND MATHEMATICAL MODEL

3.1. Hydroelectric Power Calculation

In hydroelectric energy generation, theoretical power depends on water flow rate, net head height, and overall system efficiency. The basic equation used in the calculation is:

$$P = \rho \times g \times Q \times H \times \eta$$

Where: P: Electrical power (W), ρ: Water density (1000 kg/m³), g: Gravitational acceleration (9.81 m/s²), Q: Water flow rate passing through turbine (m³/s), H: Net head height (m), η: Total system efficiency (%85). This equation represents the fundamental calculation method for converting hydraulic energy into electrical energy.

3.2. Ecological Flow Release

In hydroelectric power plants, using all of the water for energy generation is not possible in every situation. To protect the river ecosystem, a certain amount of water must be released into the natural riverbed. In this study, 10% of the inflow was accepted as ecological flow release.

$$Q_{available} = Q_{inflow} - Q_{ecological} \quad | \quad Q_{turbine} = \min(Q_{available}, Q_{maximum})$$

3.3. Calculation of Hydraulic Losses

As water moves inside penstocks, it suffers energy loss due to friction with the pipe surface and minor losses from connection fittings. The Darcy-Weisbach approach was utilized in calculating these losses:

$$h_f = f \times (L / D) \times (v^2 / 2g) \quad , \quad h_m = K \times (v^2 / 2g) \quad , \quad h_{total} = h_f + h_m$$

3.4. Net Head Calculation

The value that should be taken as the basis for power generation in hydroelectric power plants is not the gross head, but the net head obtained after subtracting hydraulic losses: $H_{net} = H_{gross} - h_{total}$. Therefore, changes in pipe diameter, water velocity, and flow rate directly affect the net head value.

4. PARAMETERS USED IN SIMULATION

In the initial baseline scenario of the model, the following design and operational parameters were used:

Parameter	Assumed Value
Gross Head	100 m
Total System Efficiency	85%
Ecological Flow Release	10% of Inflow
Maximum Turbine Flow Rate	20 m³/s
Penstock Length / Diameter	1000 m / 3.0 m
Darcy Friction Factor / Minor Loss Coeff.	0.02 / 2.0
Water Density / Gravity Acceleration	1000 kg/m³ / 9.81 m/s²
Monthly Inflow Rates (Jan – Dec)	12, 14, 18, 25, 30, 28, 22, 16, 12, 10, 9, 11 m³/s

Important note: These flow rates are hypothetical values created to run the model and analyze operating conditions, not actual measured data.

5. METHODOLOGY AND ANALYSES PERFORMED

The project was developed in Python and executed on Google Colab using NumPy, Pandas, and Matplotlib. Monthly flow usable by turbines, net head, hydraulic losses, and total yearly energy outputs were computed across varying penstock diameters (2.5m to 4.5m) and maximum turbine flow rates (12 m³/s to 24 m³/s).

6. OBTAINED RESULTS

6.1. Baseline Operating Scenario Results

Result	Value
Annual Total Energy Generation (MWh)	102,055.61 MWh
Annual Total Energy Generation (GWh)	102.056 GWh
Annual Average Power Output	11.650 MW
Time-Weighted Average Hydraulic Loss	1.991 m
Time-Weighted Average Net Head	98.009 m

6.2. Design Parameter Analysis Results

In the design sweep across penstock diameter and turbine flow combinations, the highest annual energy generation was achieved under: **Penstock Diameter: 4.5 m | Max Turbine Flow: 24 m³/s | Annual Energy Generation: 110.426 GWh**.

7. ENGINEERING EVALUATION

First, a high water flow rate does not always mean higher electricity generation in the same ratio because ecological flow (10%) and turbine capacity limit usable flow. Second, hydraulic losses directly impact power yield; thus penstock diameter and flow velocity are critical design parameters. Third, evaluation requires looking at year-round available water, not just peak turbine capacity.

8. LIMITATIONS AND FUTURE IMPROVEMENTS

Limitations include hypothetical inflow data, fixed 85% efficiency, simplified Darcy-Weisbach losses, lack of detailed turbine partial-load curves, and exclusion of transient phenomena (water hammer, cavitation) or economic cost optimizations.

9. CONCLUSION

A numerical simulation model was successfully implemented using Python and Google Colab. For the baseline scenario, annual generation was calculated as 102.055,61 MWh (102.056 GWh) with an average power of 11.650 MW. In design optimization, a peak of 110.426 GWh/year was attained. The results emphasize the importance of reducing hydraulic losses, selecting appropriate turbine capacity, and managing water resources accurately.